T-150 UAS Plugin Interaction Architecture

6DOF Mover (Physics)
Battery Processor
Propulsion Processor
VRS Processor
Slung Load Processor
AuxData Layer

AFSIM SIMULATION ENGINE

ROTORCRAFT_6DOF_MOVER

(Physics Engine)

• Rigid Body 6DOF • 8 Rotor Models

• Flight Controllers • Quaternion Attitude

20 Hz

6dof_total_power
6dof_vel_*
6dof_pos_*

INPUTS

PLATFORM AUXDATA (Key-Value Communication Layer)

6dof_total_power | 6dof_vel_* | 6dof_pos_* | Platform Altitude | Platform Accel

BATTERY PROCESSOR

• Li-Ion Chemistry

• Voltage Sag Model

• SOC Tracking

PROPULSION PROCESSOR

• Motor Thermal

• Ground Effect

• Payload Mass

VRS PROCESSOR

• Descent Monitor

• 5-State Machine

• Lift Degradation

SLUNG LOAD PROCESSOR

• Pendulum Dynamics

• Cable Tension

• Resonance Detect

10 Hz

(50 Hz for Slung Load)

OUTPUTS

AUXDATA OUTPUTS (Available to Scenario Scripts)

BATTERY_SOC | PROP_TOTAL_POWER | VRS_MARGIN | SLUNG_TENSION | + 50 more keys